

Damien Robert

Deep Learning for Remote Sensing & Environment



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I am a postdoctoral researcher in the [EcoVision](#) lab at the [University of Zurich](#), where I collaborate with [Jan Dirk Wegner](#) to design deep learning methods for remote sensing and environmental applications. Before joining UZH, I completed my PhD on “*Efficient Learning on Large-Scale 3D Point Clouds*” at [IGN](#) and [ENGIE](#), under the supervision of [Loic Landrieu](#) and [Bruno Vallet](#).

■ POSITIONS

2024–now	Postdoctoral Researcher EcoVision lab, University of Zurich · Zurich, Switzerland Deep learning for remote sensing and environment PI: Jan Dirk Wegner
2020–2024	PhD Student LASTIG, IGN / ENGIE Lab CRIGEN · Paris, France Efficient learning on large-scale 3D point clouds Advisors: Loic Landrieu , Bruno Vallet
2017–2020	R&D Engineer SIRADEL, ENGIE / ENGIE · Rennes, France Deep learning on large-scale terrestrial/aerial, indoor/outdoor 3D and 2D data
2017	Co-Founder Inspirama · Rennes, France Website gathering book recommendations from inspiring people
2015	R&D Intern Dassault Systèmes · Providence, RI, USA Dimensionality reduction and dynamic system modeling
2014	R&D Intern Dassault Systèmes · Providence, RI, USA UX design

■ EDUCATION

2020–2024	PhD, Computer Science Université Gustave Eiffel · Paris, France Efficient Learning on Large-Scale 3D Point Clouds
2022	Summer School ICVSS · Sicily, Italy Computer vision courses by world-renowned experts in academia and industry
2017	AI Fall School CNRS · Lyon, France Multi-disciplinary course for AI students and researchers
2011–2015	MSc, Engineering École Centrale de Lyon · Lyon, France Mathematics, Computer Science, Mechanics, Signal Processing, Automation

■ AWARDS

2026	Outstanding Reviewer · CVPR
2025	PhD Award, accessit (runner-up) · AFRIF French Association for Pattern Recognition and Interpretation
2024	Outstanding Reviewer · ECCV

2026

Toward a Foundation Model for Forest Point Clouds

Y. Yue, S. Puliti, D. Robert, A. Topaloğlu, B. Xiang, M. Wielgosz, J. D. Wegner, R. Astrup, C. Rupprecht, K. Schindler
Under review

From Machine Learning to Large-Scale EO Products: Best Practices for Making Maps

G. Sialelli, R. Young, Y. Jiang, C. Aybar, L. Scheibenreif, D. Robert, C. Mosig, M. Czerkawski, A. J. Stewart, J. D. Wegner, A. Pirinen, O. Mogren, K. Schindler
Under review

Bias Aware Benchmark for Species Distribution Modeling

E. Arens, N. van Tiel, R. Zbinden, C. Vanalli, D. Robert, L. Drees, B. Kellenberger, L. Pellissier, J. D. Wegner, D. Tuia
[World Biodiversity Forum](#)

LitePT: Lighter Yet Stronger Point Transformer

Y. Yue, D. Robert, J. Wang, S. Hong, J. D. Wegner, C. Rupprecht, K. Schindler
 CVPR [Highlight](#)

EZ-SP: Fast and Lightweight Superpoint-Based 3D Segmentation

L. Geist, L. Landrieu, D. Robert
[ICRA](#)

FORMSpOT: Revealing fine-scale forest disturbances from nation-wide 1.5 m forest canopy height time series

M. Schwartz, F. Fogel, N. Besic, D. Robert, L. Geist, J. Renaud, J. Monnet, C. Mosig, C. Vega, A. d'Aspremont, L. Landrieu, P. Ciais
[Remote Sensing of Environment](#)

2025

MT-GSR4B: Multi-Temporally Guided Super-Resolution for Above-Ground Biomass Estimation

K. Karaman, V. S. F. Garnot, D. Robert, M. J. Santos, J. D. Wegner
[PFG – Journal of Photogrammetry, Remote Sensing and Geoinformation Science](#)

SSL4Eco: A Global Seasonal Dataset for Geospatial Foundation Models in Ecology

E. Plekhanova, D. Robert, J. Dollinger, E. Arens, P. Brun, J. D. Wegner, N. Zimmermann
[CVPR workshop EarthVision](#)

GSR4B: Biomass Map Super-Resolution with Sentinel-1/2 Guidance

K. Karaman, Y. Jiang, D. Robert, V. S. F. Garnot, M. J. Santos, J. D. Wegner
[ISPRS Geospatial Week](#) [Oral](#)

Climplicit: Climatic Implicit Embeddings for Global Ecological Tasks

J. Dollinger, D. Robert, E. Plekhanova, L. Drees, J. D. Wegner
[ICLR workshop Tackling Climate Change with Machine Learning](#)

Lossy Neural Compression for Geospatial Analytics: A Review

C. Gomes, I. Wittmann, D. Robert, J. Jakubik, T. Reichelt, M. Martone, S. Maurogiovanni, R. Vinge, J. Hurst, E. Scheurer, R. Sedona, T. Brunschweiler, S. Kesselheim, M. Batic, P. Stier, J. D. Wegner, G. Cavallaro, E. Pebesma, M. Marszałek, M. A. B. Plomer, K. Adriko, P. Fraccaro, R. Kienzler, R. Briq, S. Benassou, M. Lazzarini, C. M. Albrecht
[IEEE Geoscience and Remote Sensing Magazine](#)

2024

Efficient Learning on Large-Scale 3D Point Clouds

D. Robert
[PhD Thesis, Université Gustave Eiffel](#) [AFRIF PhD award \(accessit\)](#)

Jury: Sébastien Lefèvre, Cédric Demonceaux, Patrick Pérez, Siyu Tang, Duygu Ceylan, Loic Landrieu, Bruno Vallet

Scalable 3D Panoptic Segmentation as Superpoint Graph Clustering

D. Robert, H. Raguet, L. Landrieu
[3DV](#) [Oral](#) · oral, top 5.3% of submissions

2023

Efficient 3D Semantic Segmentation with Superpoint Transformer

D. Robert, H. Raguet, L. Landrieu
[ICCV](#) · 26.8% acceptance rate

2022

Learning Multi-View Aggregation In the Wild for Large-Scale 3D Semantic Segmentation

D. Robert, B. Vallet, L. Landrieu
 CVPR [Best paper finalist](#) [Oral](#) · best-paper finalist, top 0.4% of submissions

■ COLLABORATION & SUPERVISION

Collaborators	Konrad Schindler ETH · Christian Rupprecht Oxford · Jan Dirk Wegner UZH · Loic Landrieu ENPC · Vivien Sainte Fare Garnot UZH · Lukas Drees UZH · Martin Schwartz LSCE · Elena Plekhanova WSL
PhD Students	Emilia Arens UZH with Jan Dirk Wegner · Johannes Dollinger UZH with Jan Dirk Wegner · Kaan Karaman UZH with Jan Dirk Wegner · Louis Geist ENPC with Loic Landrieu · Yuanwen Yue ETH with Konrad Schindler and Christian Rupprecht
Master Students	Valerio Schelbert WSL · Jackson Sunny École Polytechnique with SAMP
Advising	UpCircle AI Switzerland · École des Sciences Criminelles de Lausanne Switzerland · SAMP AI France · GEODIGITAL US

■ REVIEWING, CHAIRING & ORGANISING

 Reviewer  Area chair  Organising committee

2027	 CVPR workshop EarthVision
2026	 ECCV ·  CVPR
2025	 CVPR ·  CVPR workshop EarthVision
2024	 ECCV ·  ISPRS Journal of Photogrammetry and Remote Sensing ·  ICLR workshop ML4RS ·  CVPR workshop EarthVision
2023	 CVPR ·  CVPR workshop EarthVision
2022	 ISPRS Journal of Photogrammetry and Remote Sensing ·  CVPR workshop EarthVision

■ TEACHING

2026	Transformer · UZH · Lecture, Master 2
2025	AI4Good group projects · UZH · Labs, Master 2
2025	Transformer · UZH · Lecture, Master 2
2024	Transformer, NeRFs, and Diffusion · UZH · Lecture and labs, Master 2
2023	Deep Learning for Remote Sensing · ENSG-IGN · Lecture, Master 2
2022	3D Deep Learning for Remote Sensing · ISPRS'22 · Tutorial, Researchers
2022	3D Deep Learning, Torch-Points3D & DeepViewAgg · ENGIE CRIGEN lab · Tutorial, Researchers
2022	Deep Learning for Remote Sensing · ENSG-IGN · Lecture, Master 2
2020	Deep Learning for Computer Vision · École Polytechnique · Lecture, Master 1

■ OPEN-SOURCE SOFTWARE

★ 1055  136	Superpoint Transformer ·  drprojects/superpoint_transformer
★ 406  45	LitePT ·  prs-eth/LitePT
★ 237  24	DeepViewAgg ·  drprojects/DeepViewAgg
★ 74  8	Point Geometric Features ·  drprojects/point_geometric_features
★ 53  3	NoRA ·  drprojects/nora
★ 15  0	torch-graph-components ·  drprojects/torch-graph-components

■ SELECTED TALKS

2026	ESA-NASA workshop on AI Foundation Model for EO · tutorial · Building Earth Observation Embedding Workflows with TerraTorch · Virtual
	ESA Φ-lab · Geospatial foundation models for forest disturbance monitoring · Frascati, Italy
2025	MPI of Animal Behavior · Research at EcoVision lab · Zurich, Switzerland
	Global Forest Watch · Geospatial foundation models for forest disturbance monitoring · Virtual
	NatureFinance · Research at EcoVision lab · Virtual

	NVIDIA Strategic Research Engagement · Superpoint Transformer and EcoVision research · Zurich, Switzerland
	CNES COMET TSI · Remote Sensing for Species Distribution Modeling · Virtual
2024	3D Data Academy · interview · Interview by Florent Poux and tutorial for Superpoint Transformer · YouTube
	Google DeepMind · Research at EcoVision lab · Zurich, Switzerland
	3DV'24 · oral · Scalable 3D Panoptic Segmentation as Superpoint Graph Clustering · Davos, Switzerland
2023	École des Ponts, IMAGINE lab · Efficient Learning on Large-Scale 3D Point Clouds · Paris, France
	National Land Survey of Finland (NLS) · IGN's research on Large-Scale 2D and 3D Learning · Paris, France
	ETH Zürich, Computer Vision and Geometry lab · Efficient Learning on Large-Scale 3D Point Clouds · Zurich, Switzerland
	ETH Zürich, Photogrammetry and Remote Sensing lab · Efficient Learning on Large-Scale 3D Point Clouds · Zurich, Switzerland
	Valeo.ai · Efficient 3D Semantic Segmentation with Superpoint Transformer · Paris, France
2022	CV News · interview · Interviewed by the CV News journal for its Best of CVPR'22 issue · New Orleans, US
	CVPR'22 · oral · Learning Multi-View Aggregation In the Wild for Large-Scale 3D Semantic Segmentation · New Orleans, US
	45 talks given since 2021 — full list at drprojects.github.io/talks .

SKILLS & LANGUAGES

Research	3D & geometry Large-scale 3D point clouds · LiDAR data · Superpoint-based learning · Graph-based learning and graph partitioning			
	Efficient & scalable learning Efficient deep learning · Scalable model design · EO data representation and compression			
	Learning paradigms Multimodal learning · Self-supervised learning · Representation learning			
	Environment & Earth observation Remote sensing · Forest structure analysis · Species distribution modeling · Representation learning for macroecology			
Tools	 Python	 PyTorch	 PyTorch Lightning	 PyTorch Geometric
	 C++	 Hydra	 Weights & Biases	 Git
	 Linux	SLURM	 scikit-learn	 Plotly
	 Blender	LaTeX		
Languages	French Native	English Fluent	Spanish Intermediate	German Beginner
Abroad	Zurich, Switzerland 2024–now · Backpacking, Worldwide 2015–2016 · Providence, RI, United States 2014–2015			
Interests	Nature & forests · Boxing · Mountaineering · Skateboarding · Guitar · Painting · Reading · Board games			